

.NET Conf

Focus on MAUI

August 9, 2022

focus.dotnetconf.net

Welcome to .NET Conf: Focus on MAUI

David Ortinau, Principal PM

@davidortinau

Maddy Montaquila, Senior PM

@maddymontaquila





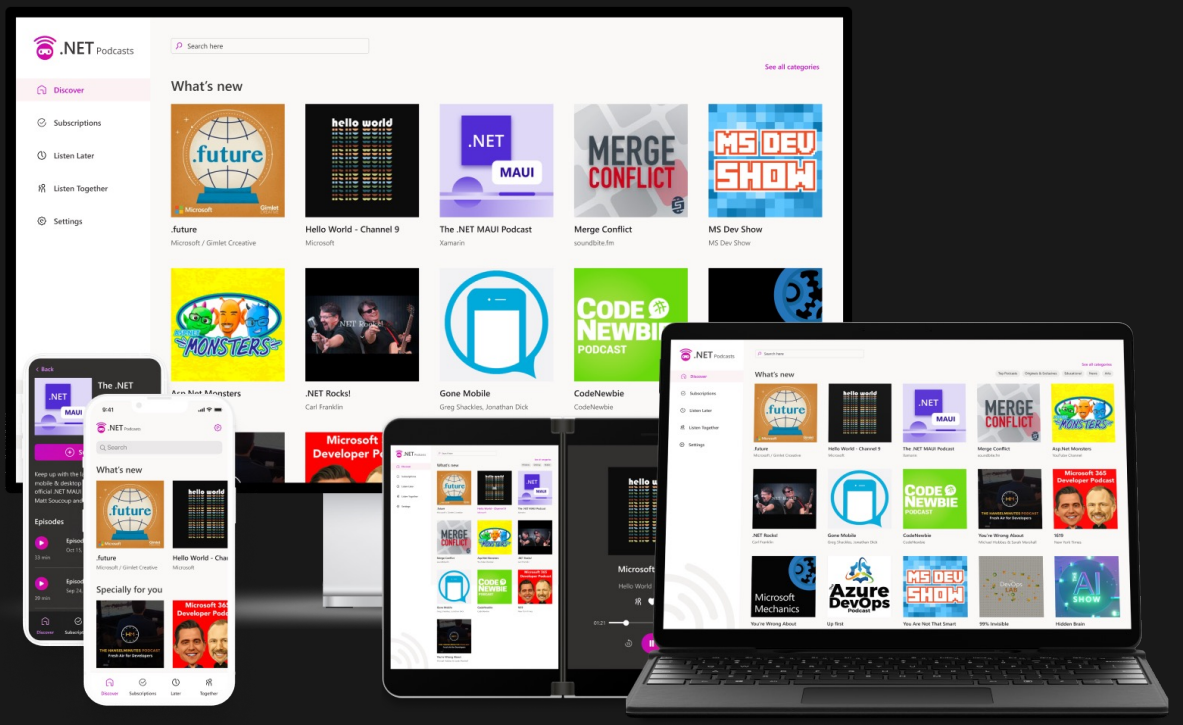
Client app development is important

>1B Windows 10 & 11 devices worldwide

>4B iPhone & Android devices worldwide

Sources: blog.google/products/android/android-12-beta/ ;
www.imore.com/apple-earnings-call-transcripts-q1-2021 ;

.NET Multi-platform App UI is the most productive way to develop native apps with a single codebase that perform great on any consumer device.



.NET MAUI in .NET 6

Unified .NET

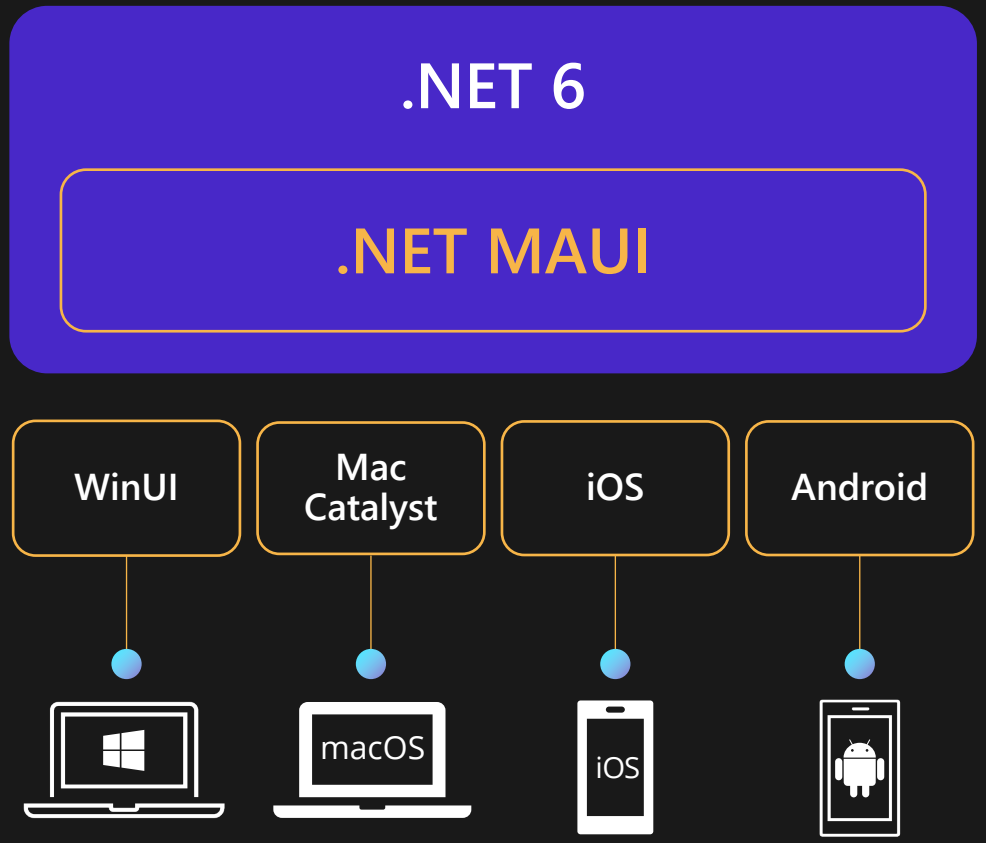
- One Base Class Library
- Type Unification
- CLI Support
- SDK Style Projects
- Modern Support Lifecycle

Mobile, desktop, hybrid

- Android
- iOS
- WinUI
- Mac Catalyst
- Blazor WebView

Productive development

- Single Project
- Microsoft.Extensions
- .NET Hot Reload
- XAML Hot Reload
- Visual Diagnostics



RELEASED

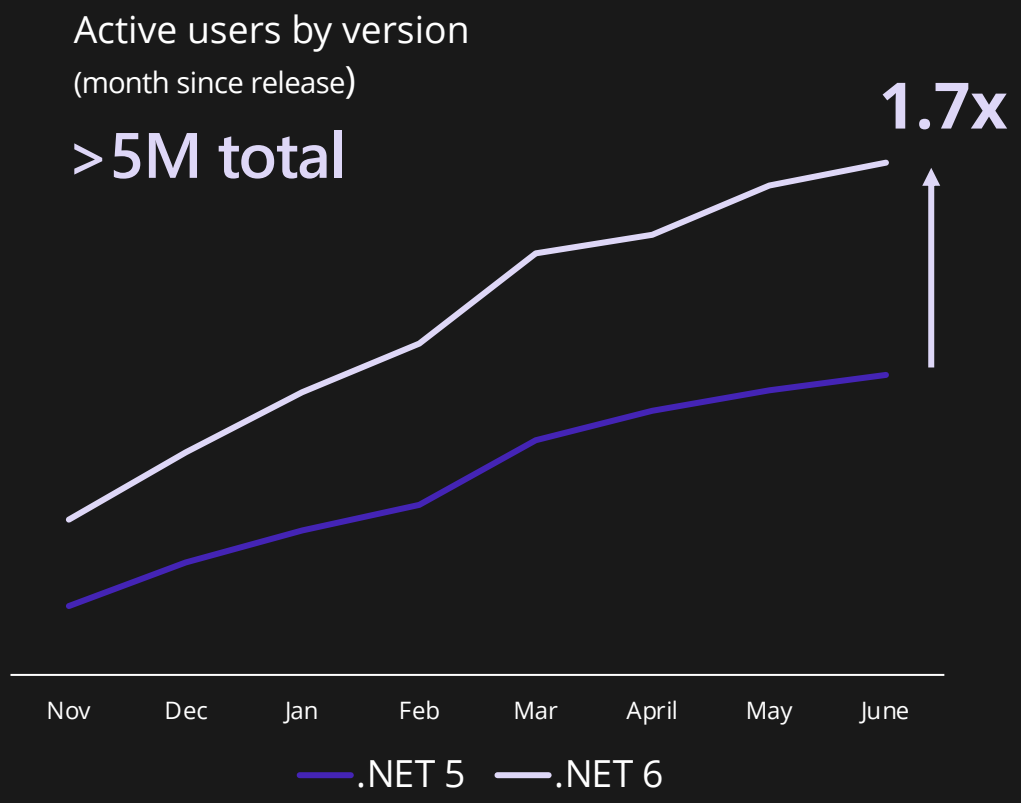
.NET Multi-platform App UI

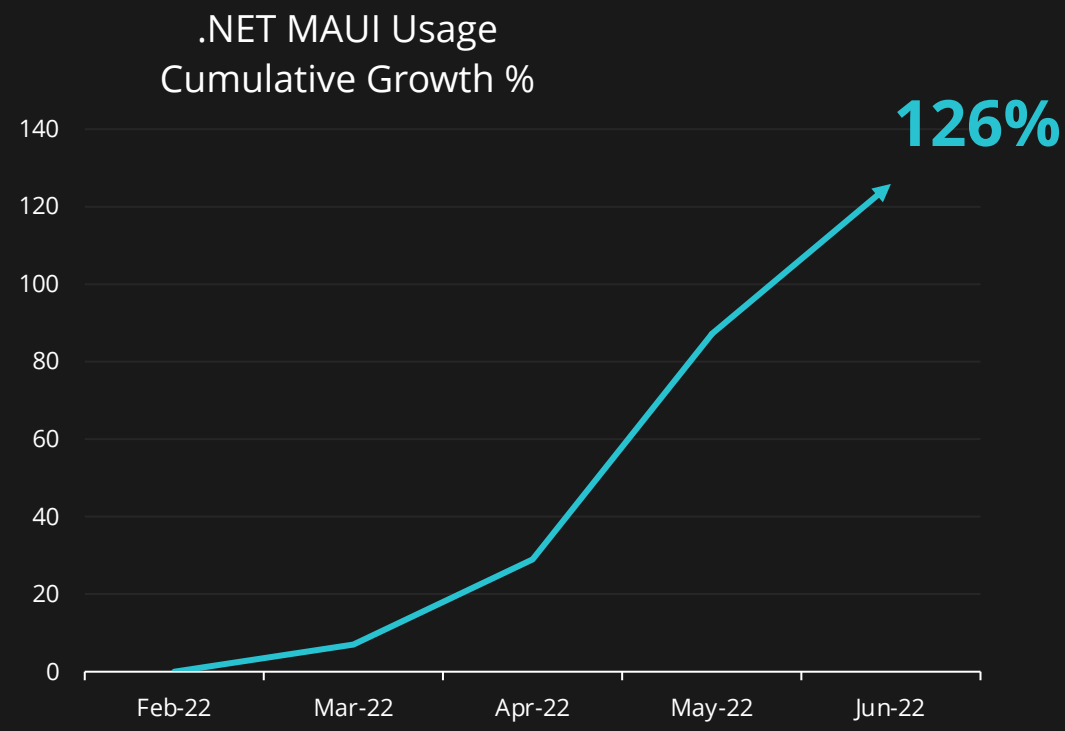
- **Today** Visual Studio 2022 17.3 released with .NET MAUI tools
- .NET MAUI SDK released in May
- Visual Studio for Mac tools will release in 17.4



Update to Visual Studio 2022 17.3 from the Visual Studio Release Channel
visualstudio.com/download

.NET Adoption





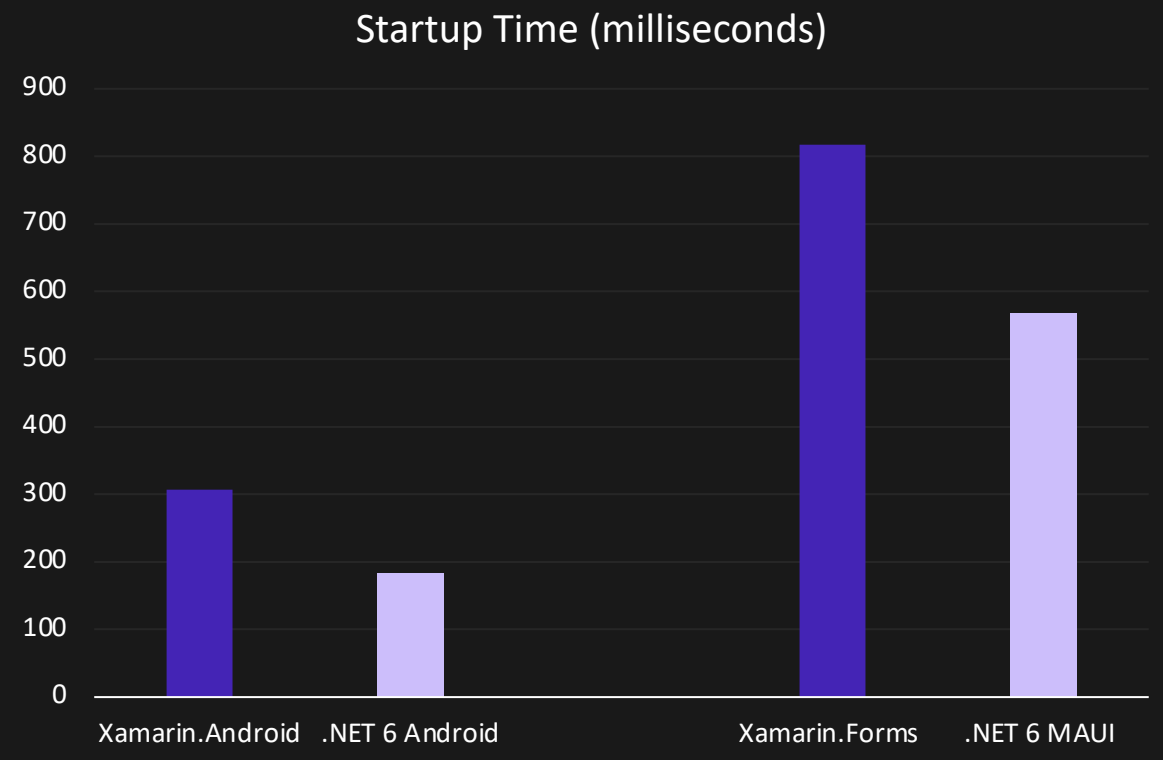
.NET MAUI perf improvements

68%

Faster than Xamarin.Android

44%

Faster than Xamarin.Forms




CHIPOTLE
MEXICAN GRILL
FORZA
HORIZON 5
stackoverflow
GoDaddy
Cincinnati Children's
changing the outcome together
GE Aviation
Alaska
Tencent 腾讯
RAYGUN
fd
SIEMENS
Healthineers
ACADEMY
OF MOTION PICTURE
ARTS AND SCIENCES
Microsoft
Dynamics 365
NATIONAL
INSTRUMENTS
Optimizely
VQ COMMUNICATION
GEICO
NetEase
Games
SETPOINT®
MEDICAL
The
POSTAGE
ASOS
American
Cancer
Society®
Microsoft
Azure
Microsoft
Bing
e>olution
SOFTWARE
ShoWorks™
LLYRIAD
Carve your empire. Change the world.
2GIS
STATSports®
LightSail
OUTBACK
STEAKHOUSE®
AstroReality
FUN
ROCK
ZERO
FRICTION

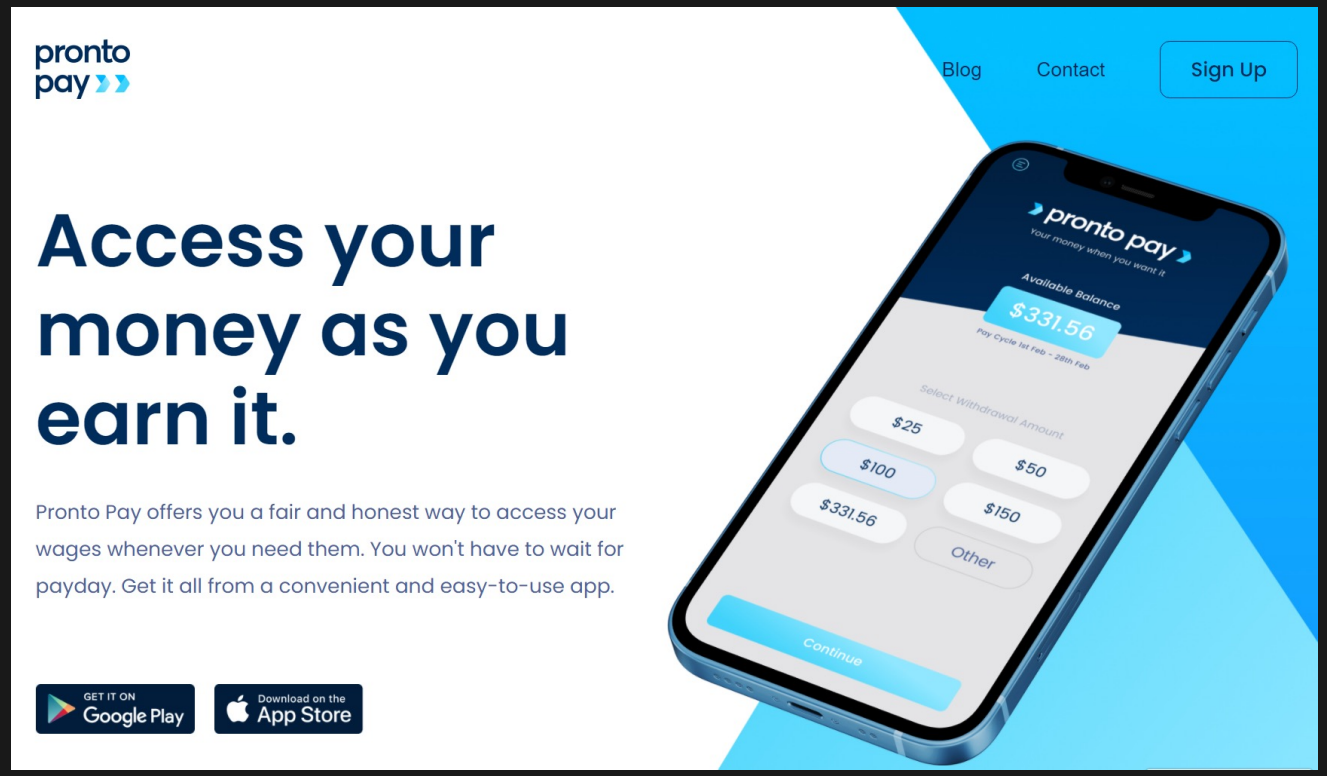
Pronto Pay

“We chose to go into production with .NET MAUI because as a startup, we need to move fast and want be on the cutting edge of tech.

We love the simplicity of a single code-base and the usability on iOS, Android, macOS, and Windows.

And with the ease of deployment built into Visual Studio, we had a tool that allowed us to build a product that was dependable for our customers, yet nimble enough to continue growth with a startup team.”

- Jeff Price, Founder & CEO



Technologies used

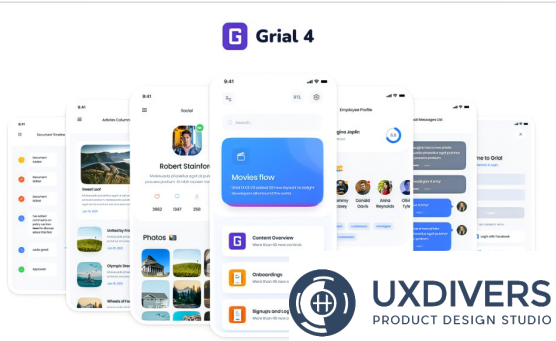
Azure, .NET MAUI, Blazor, GitHub, Visual Studio

Controls, components, plugins, libraries and more...









AndroidX
AlohaKit
CommunityToolkit.MVVM
CommunityToolkit.Maui
CommunityToolkit MauiCompat
CommunityToolkit Markup.MauiCompat
DevExpress
Facebook
FreshMvvm.Maui
Google APIs for iOS
Google Play Services Client Libraries
GrialKit
MauiAnimation
Microsoft.Maui.Graphics
MR.Gestures
MSAL
Prism.Maui
Plugin.Fingerprint
Plugin.InAppBilling
Plugin.StoreReview
Plugin.ValidationRules
Progress Telerik UI for .NET MAUI
ReactiveUI.Maui
Realm by MongoDB
Sentry
Shiny
SkiaSharp & Skottie
Syncfusion .NET MAUI Controls
TemplateUI
User Dialogs

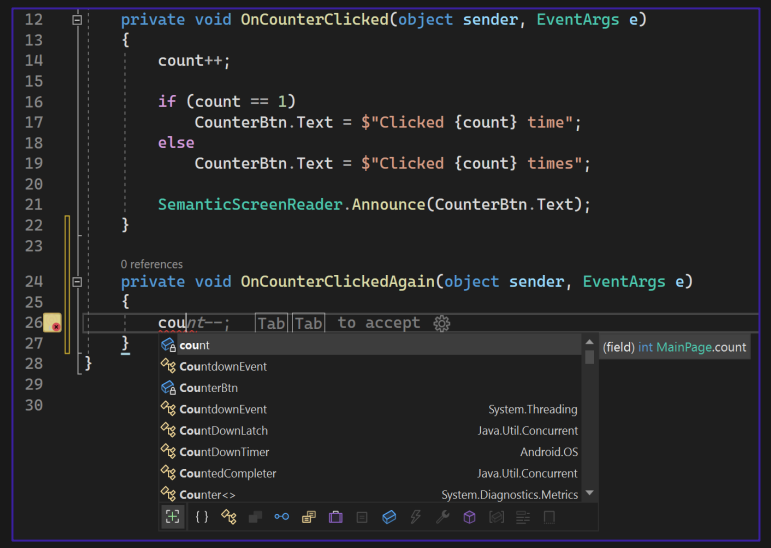

THANK YOU!

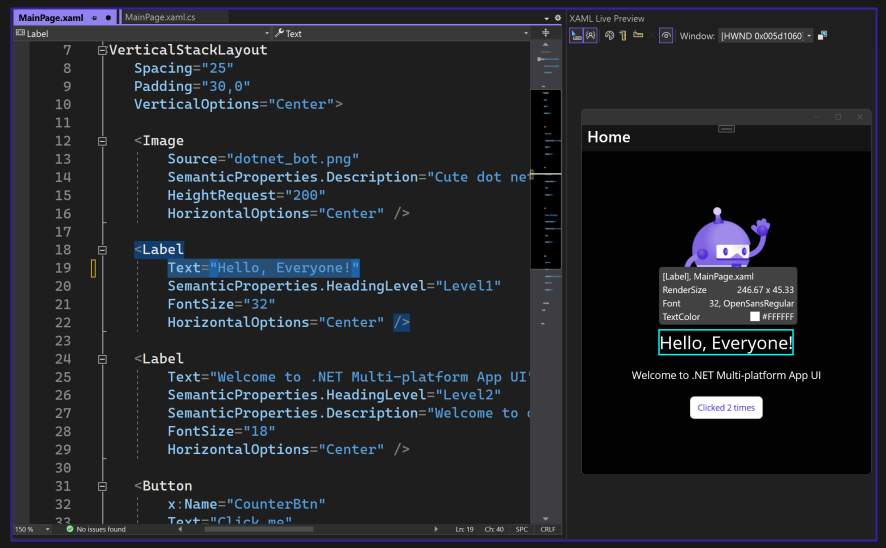
Productive

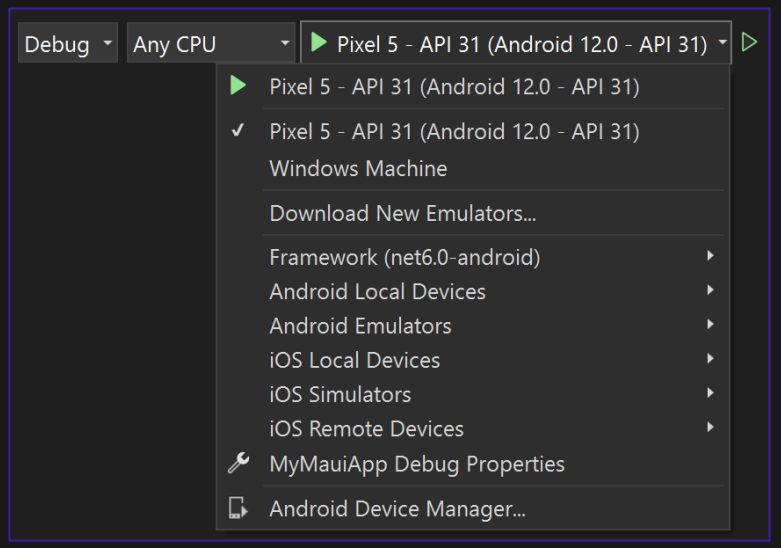
Visual Studio 2022 + .NET MAUI = Build apps faster.



Visual Studio 2022 productivity







Intelligent code editing

AI-assisted development
Live Share remote collaboration
C#, XAML, Razor IntelliSense

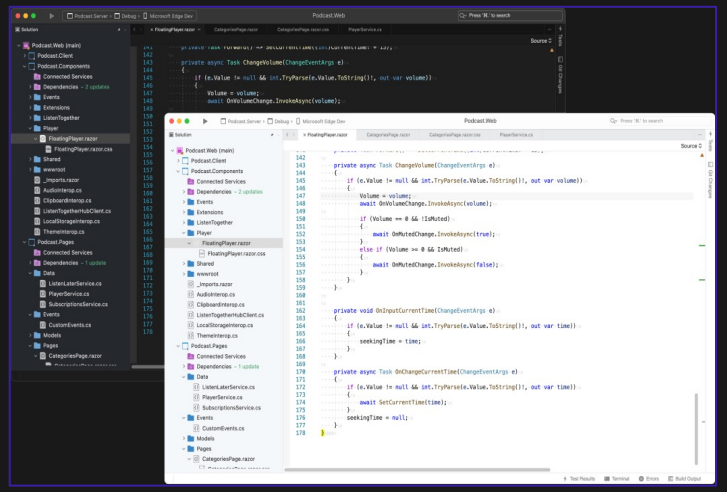
Rapid iteration

Hot reload XAML, C#, Razor, CSS, and more
Preview and Inspect UI at debug time

Develop for anything, anywhere

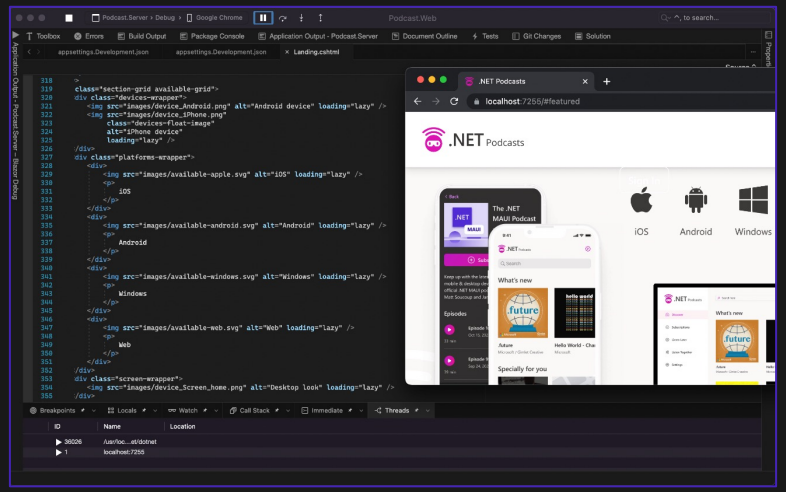
Use Android emulators, devices, or WSA
Develop iOS directly from Windows
Use iOS simulators via Xcode
Visual Studio for Mac and Windows

Visual Studio 2022 for Mac



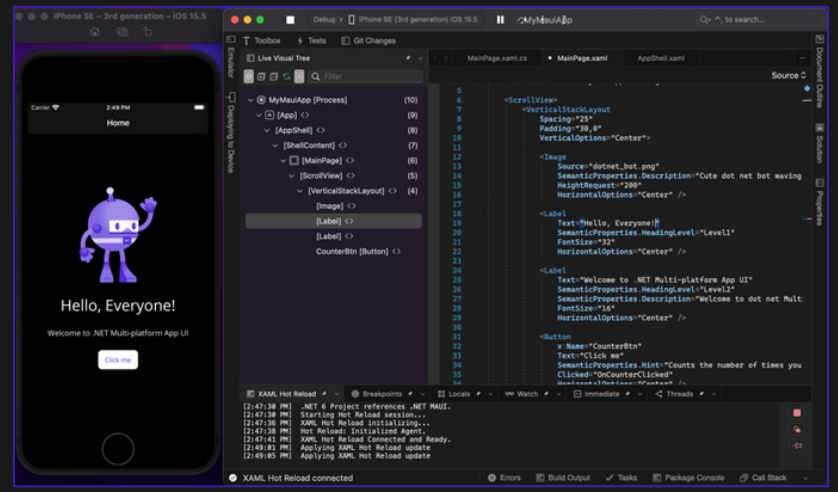
Fast & fluid IDE

Fully native macOS UI
Native support for Apple M1
Built on .NET 6



Modern .NET 6 development

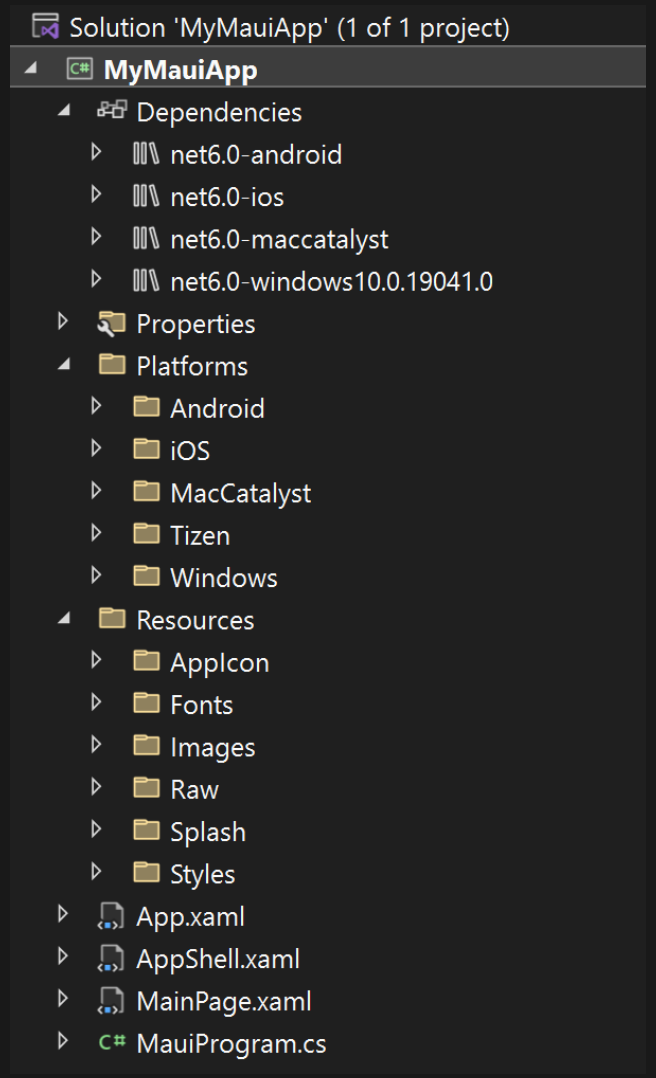
Native iOS, macOS & Android apps
Responsive web UIs
Web, cloud, & container development

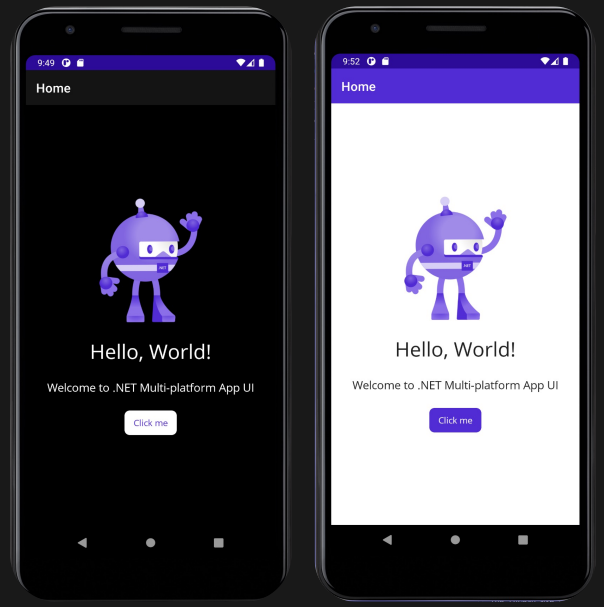


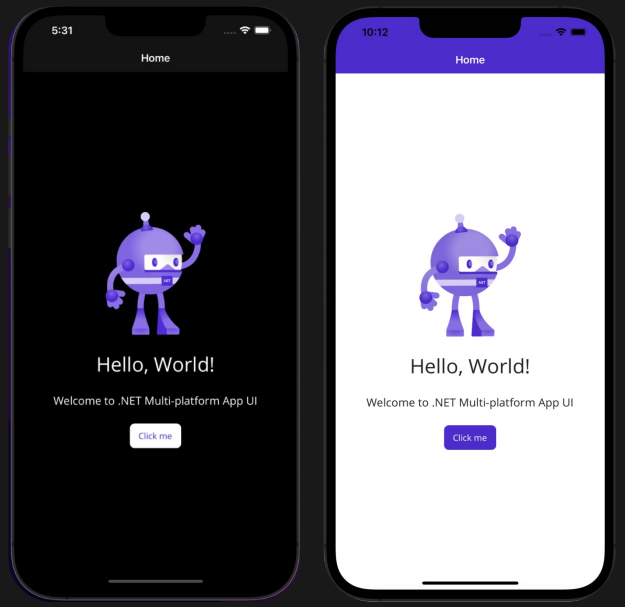
Integrated Tools

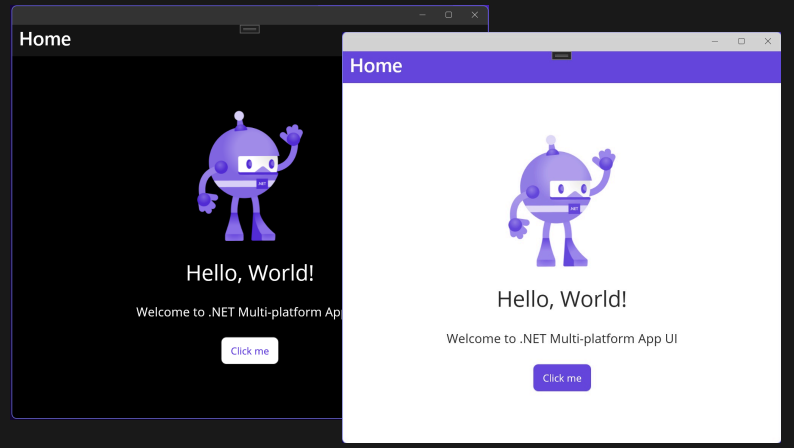
Hot reload XAML and live update UI
Fast deploy to iOS simulators & Android emulators
Powerful debugging, refactoring, IntelliSense

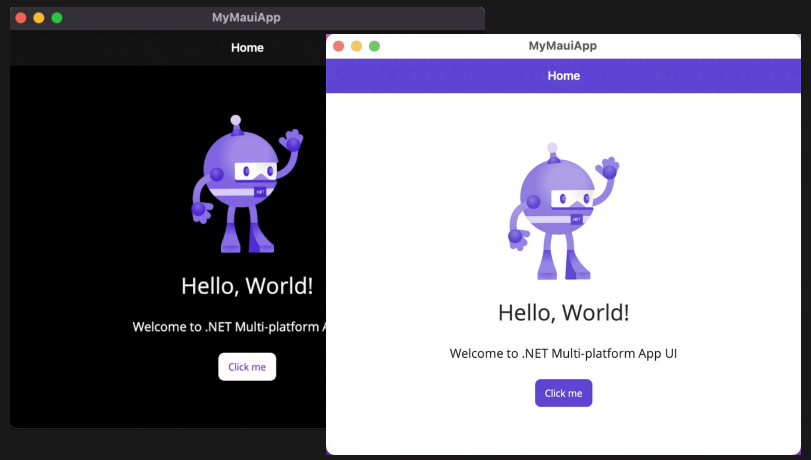
One codebase, single project, multi target









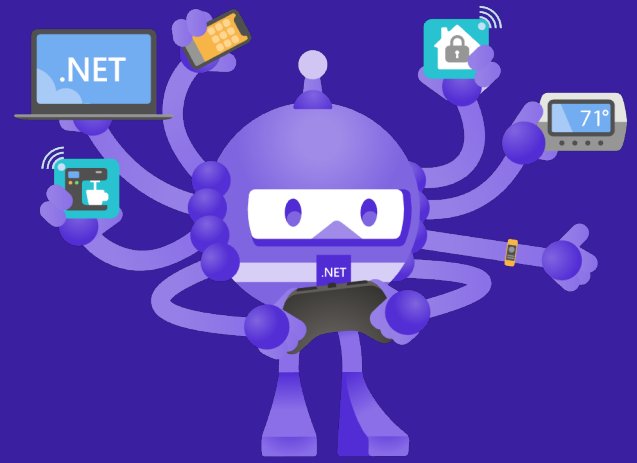


Demo

.NET MAUI Productivity

Native

Platform experiences are the best experiences.



Native



- android.widget
- AndroidX libraries
- Java/Kotlin APIs



- UIKit
- ARKit, AVKit, CarPlay, and dozens more
- Obj-C/Swift APIs



- Mac Catalyst
- UIKit + AppKit
- Dozens of frameworks
- Obj-C/Swift APIs



- Windows App SDK
- WinUI 3
- C++ APIs

What's "in the box" and nearby?

Features

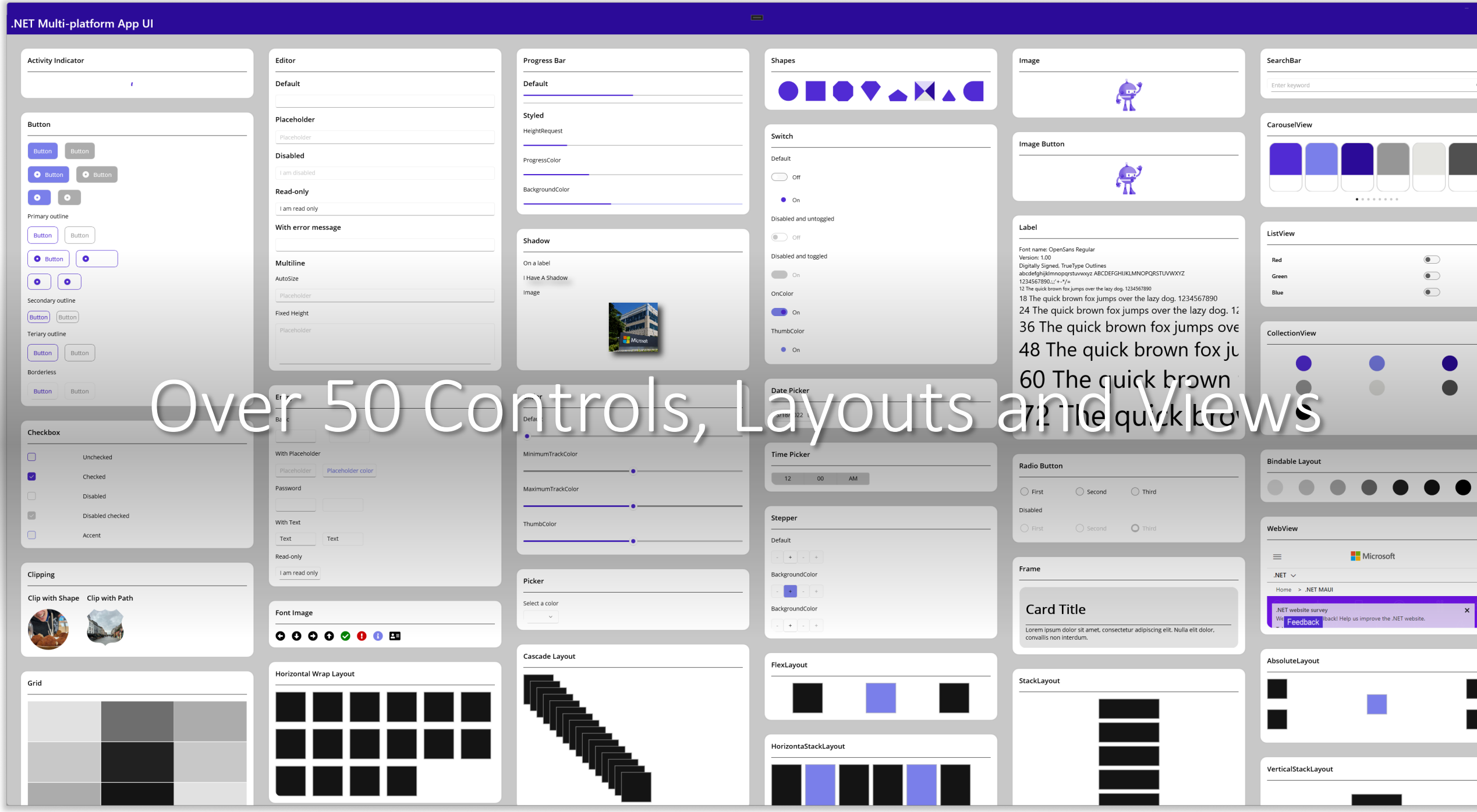
MVVM & XAML
Android, iOS, macOS, Windows
7 Layouts
44 Views
Maui.Essentials
Maui.Graphics
Xamarin.Forms Compat
CommunityToolkit.
  Maui
  Maui.Markup
  Mvvm

New Capabilities

BlazorWebView
Border
Shadows
GraphicsView
MenuBar
Window/Multi-Window
Default theme
AppIcon
SplashScreen
Fonts
Images
Shell flyout templates

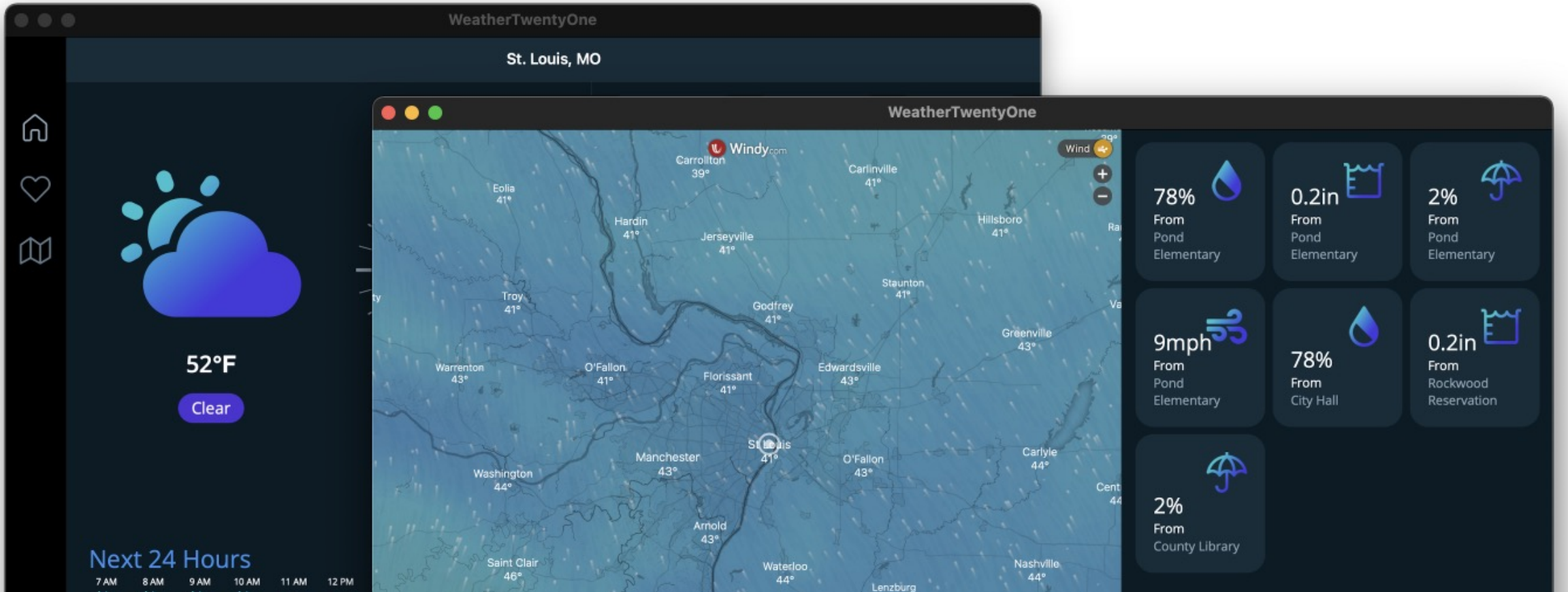
Changes from Xamarin

Color/Colors
Zero'd defaults
Namespace
More XAMLC
App Startup
Lifecycle Events
.NET workloads


.NET Multi-platform App UI
Activity Indicator
Button
Primary outline
Secondary outline
Teriary outline
Borderless
Checkbox
Unchecked
Checked
Disabled
Disabled checked
Accent
Clipping
Clip with Shape
Clip with Path
Grid
Editor
Default
Placeholder
Disabled
Read-only
I am read only
With error message
Multiline
AutoSize
Fixed Height
With Placeholder
Placeholder color
Password
With Text
Font Image
Horizontal Wrap Layout
Progress Bar
Default
Styled
HeightRequest
ProgressColor
BackgroundColor
Shadow
On a label
I Have A Shadow
Image
MinimumTrackColor
MaximumTrackColor
ThumbColor
Picker
Select a color
Cascade Layout
Shapes
Switch
Default
Off
On
Disabled and untoggled
Disabled and toggled
OnColor
ThumbColor
Date Picker
Time Picker
12
00
AM
Stepper
Default
BackgroundColor
FlexLayout
HorizontaStackLayout
Image
Image Button
Label
Font name: OpenSans Regular
Version: 1.00
Digitally Signed. TrueType Outlines
abcdefghijklmnopqrstuvwxyz ABCDEFGHIJKLMNOPQRSTUVWXYZ
1234567890.;;'+-*/=
12 The quick brown fox jumps over the lazy dog. 1234567890
18 The quick brown fox jumps over the lazy dog. 1234567890
24 The quick brown fox jumps over the lazy dog. 12
36 The quick brown fox jumps ove
48 The quick brown fox ju
60 The quick brown
72 The quick bro
Radio Button
First
Second
Third
Disabled
Frame
Card Title
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla elit dolor, convallis non interdum.
StackLayout
SearchBar
Enter keyword
CarouselView
ListView
Red
Green
Blue
CollectionView
Bindable Layout
WebView
Microsoft
.NET
Home > .NET MAUI
.NET website survey
Feedback
AbsoluteLayout
VerticalStackLayout
Over 50 Controls, Layouts and Views

Multi-window

```
var secondWindow = new Window { Page = new NewPage1 {  } };
Application.Current.OpenWindow(secondWindow);
```

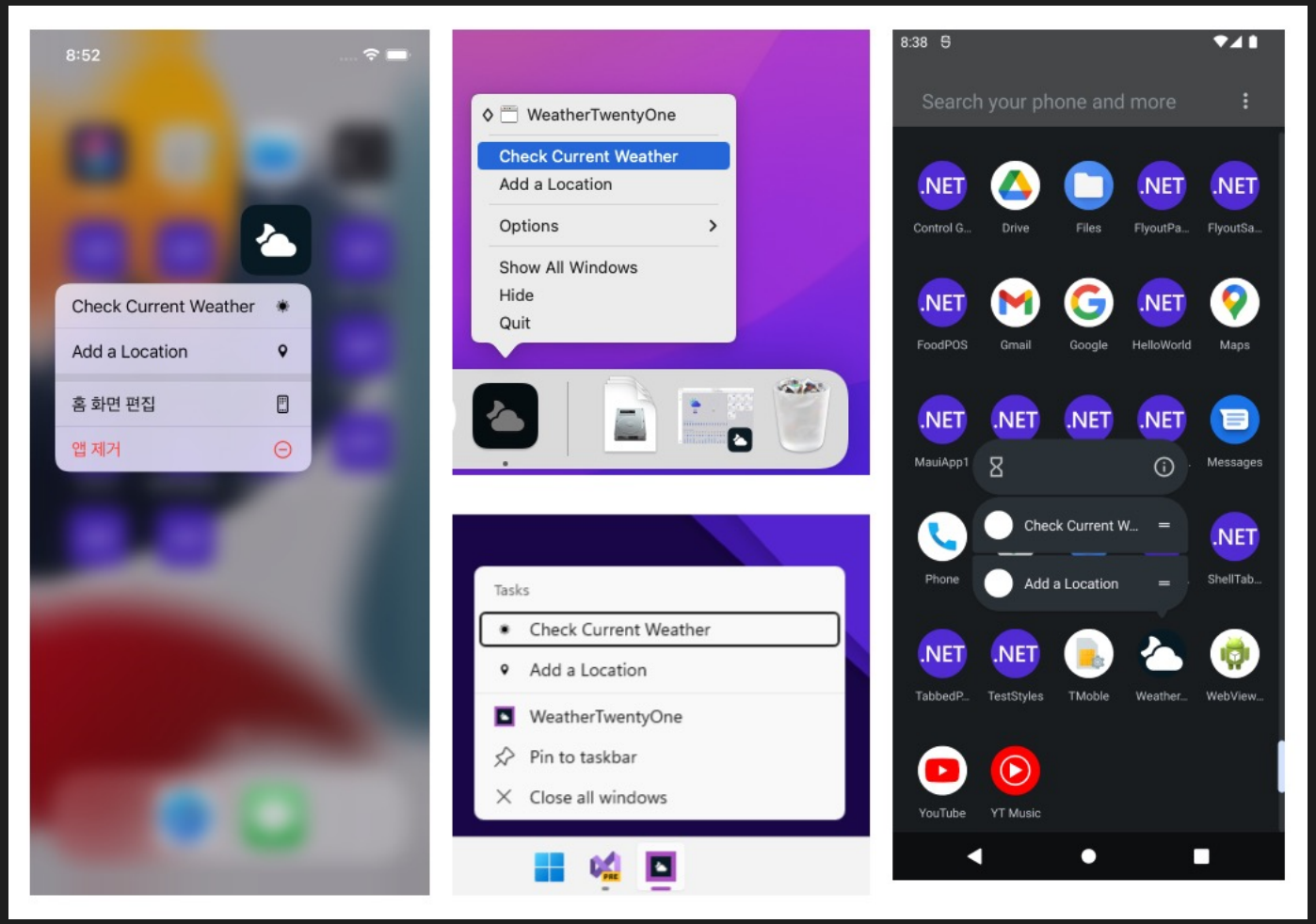


Beyond UI

Simple access to APIs

Access services and features of each platform

- App actions
- Clipboard
- Device sensors
- File system
- Network
- Notifications
- +More



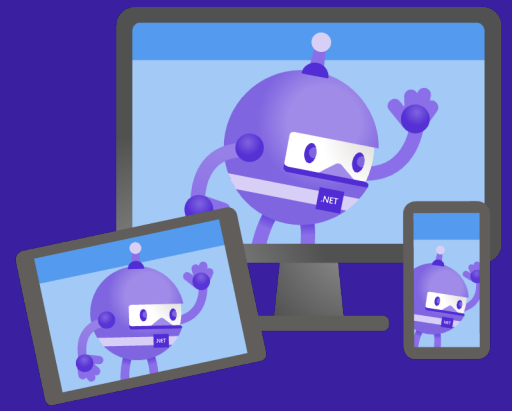

```
AppActions.SetAsync(
    new AppAction("current_info", "Check Current Weather", icon: "current_info"),
    new AppAction("add_location", "Add a Location", icon: "add_location")
    );
```

Demo

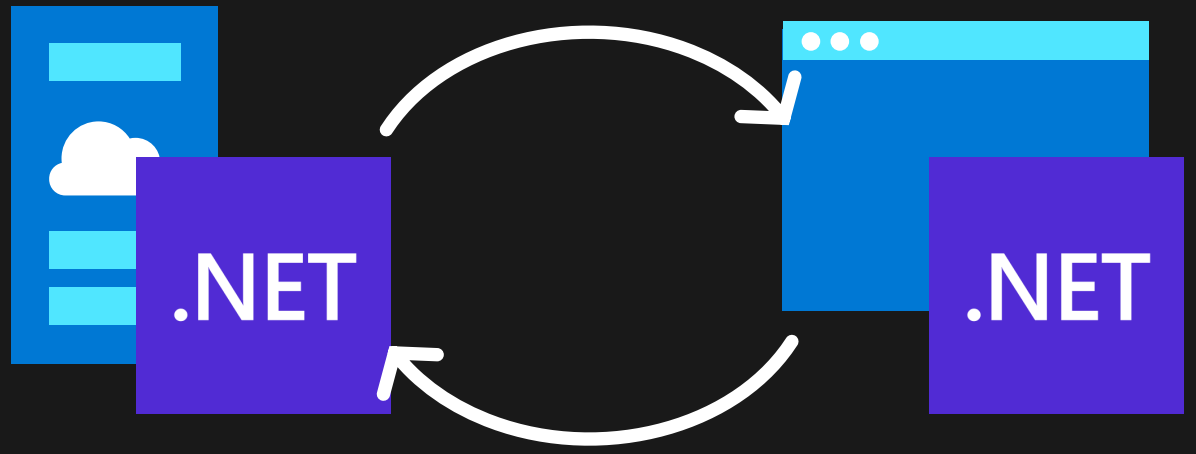
.NET MAUI Deep Dive

Hybrid

Bring native client capabilities to your web apps.



Blazor – full stack web apps with .NET



Blazor hybrid apps

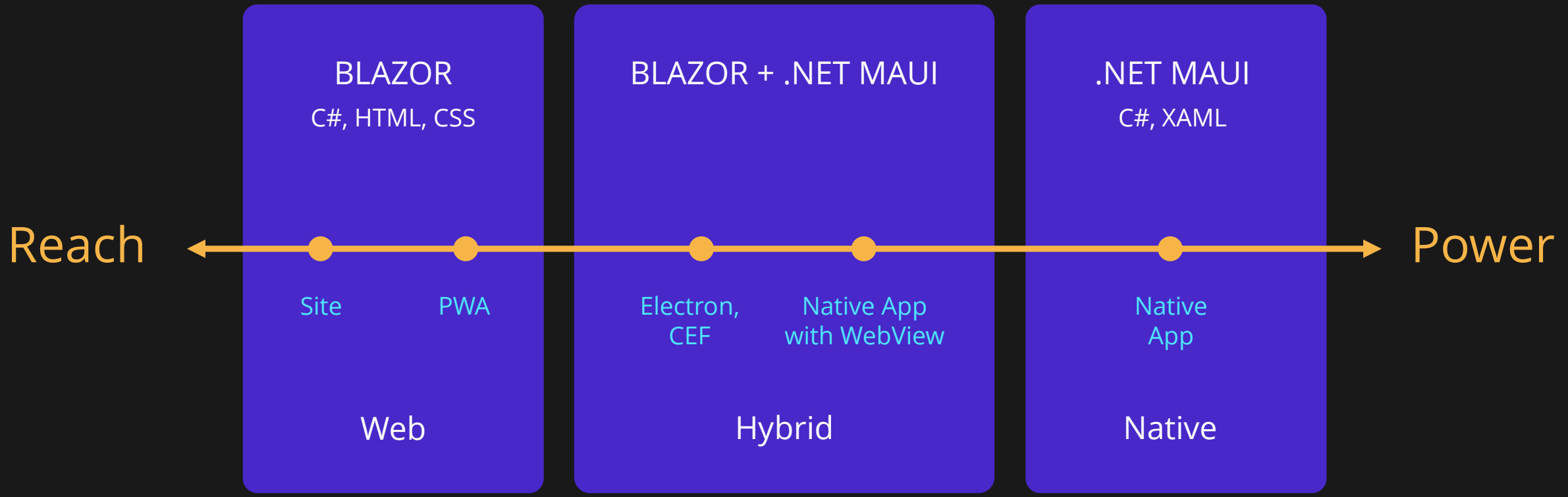
Reuse UI components across native and web

Native app container & embedded controls

Built on top of .NET Multi-platform App UI



Choose your stack



Demo

Blazor Hybrid with .NET MAUI

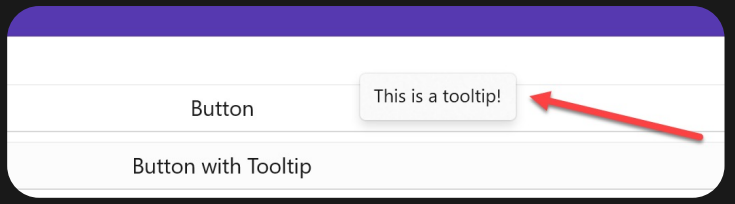
What’s next?

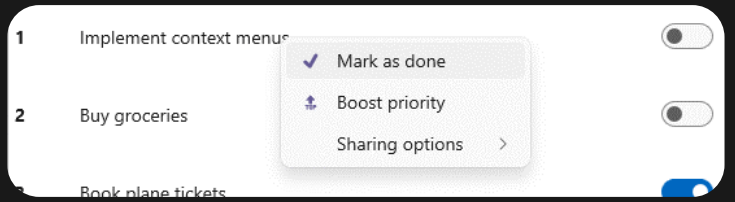


.NET 7 roadmap

.NET MAUI is great for desktop apps

ContextMenu
Cursors
Hover Gesture
Right-click Gesture
Title Bar
Tooltips



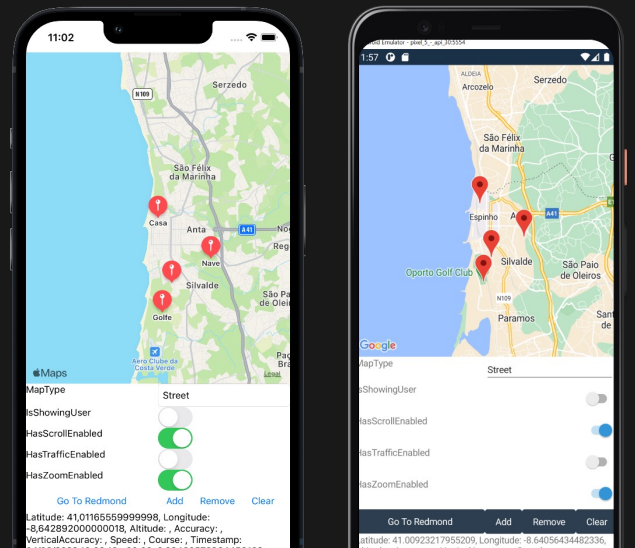


Platform interop

Channels

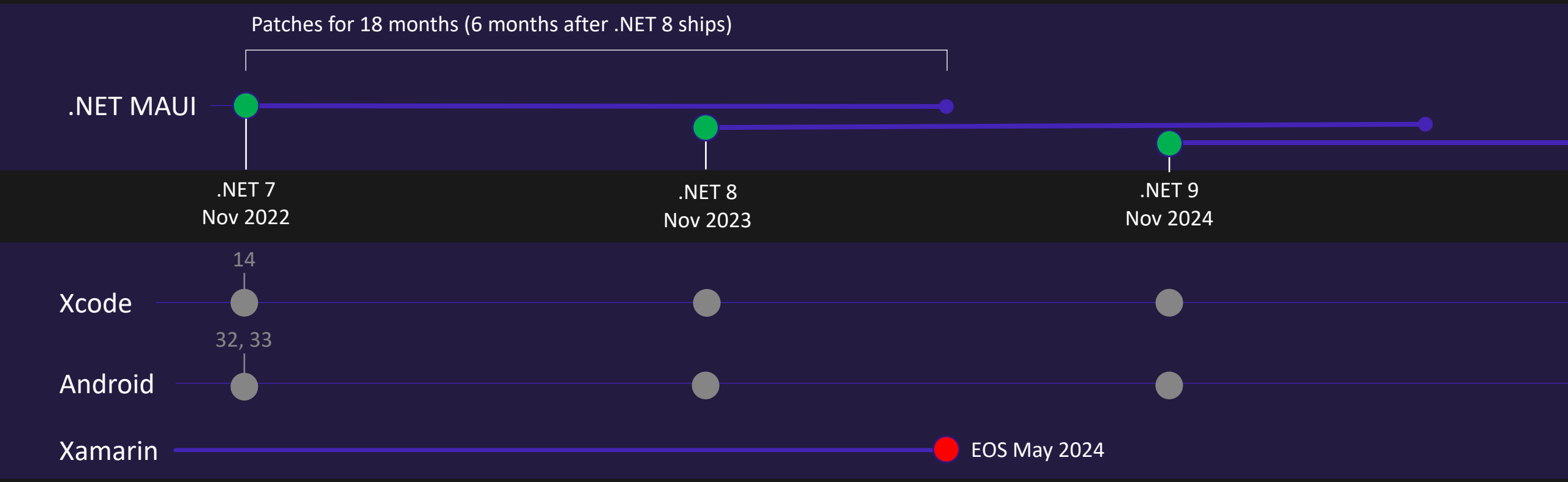
Mobile

Maps



...and more!

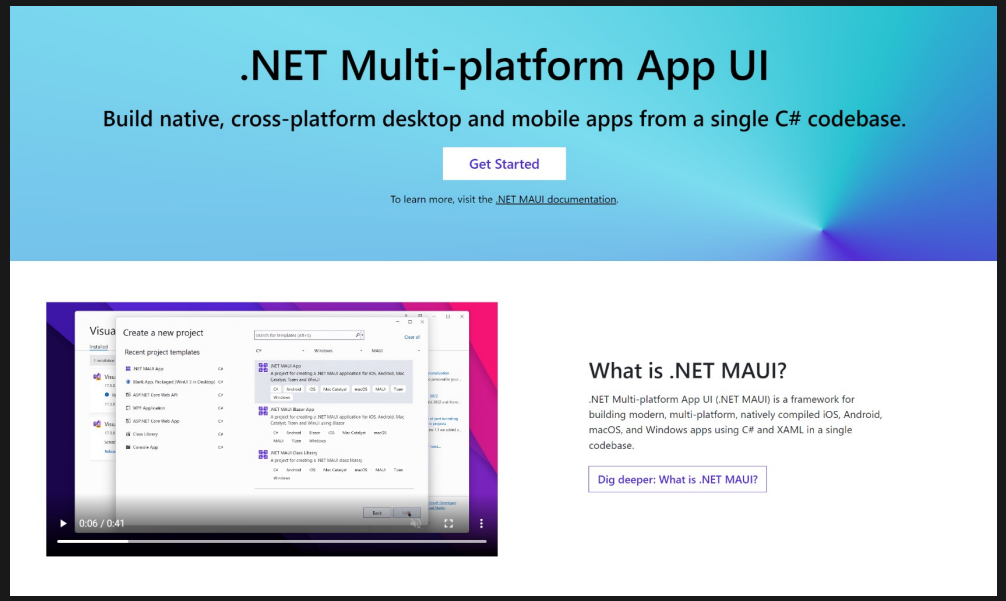
Release schedule, support & patching



- A major version of .NET MAUI will ship with .NET going forward (.NET 7, .NET 8, etc.)
- .NET MAUI is an optional workload with external dependencies (such as Xcode & Android SDK tools)
- A major version of .NET MAUI will receive patches for 6 months after a successor (major version) ships
- Xamarin SDKs will receive support until May 2024

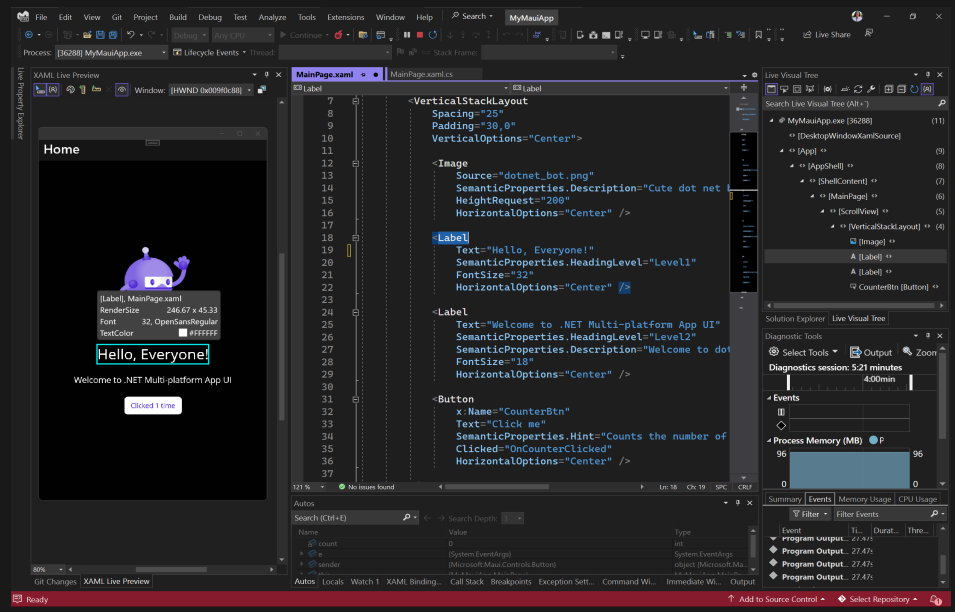
Get started today!

Get started with .NET MAUI



dot.net/maui

Download Visual Studio 2022 17.3



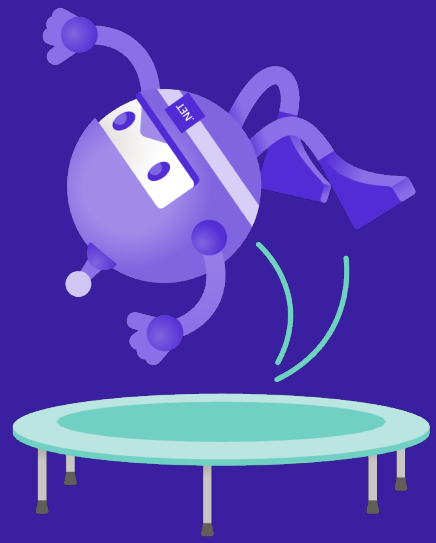
visualstudio.com/download

We want to hear from you❤️

Please take our survey and tell us your feedback. We’re listening! 👂

aka.ms/maui/survey

Ready to have fun?



.NET MAUI Coding Challenges 😎

Cloud Skills Challenge

Great for beginners, get started learning .NET MAUI and participate in some friendly competition!

aka.ms/maui/CloudChallenge

Beautiful UI Challenge

Show off your UI skills by building a beautiful .NET MAUI app!

aka.ms/maui/UIChallenge



Take a challenge and get free stickers!



Learn Live: Build mobile and desktop apps with .NET MAUI



aka.ms/maui/learnlive

EVERY WEEK STARTING SEPTEMBER 7, 2022 THROUGH NOVEMBER 16, 2022
(7 EPISODES)

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